

Matrix Algebra

1. Definition and Special Types of Matrices

Definition

- A **matrix** is a rectangular array of numbers arranged in rows and columns.

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

- Size of matrix:** $m \times n$ (m rows, n columns)
- Entries** are denoted as a_{ij} , where i is the row index and j is the column index.

Special Types of Matrices

- Vectors:**

- A **column vector** is an $n \times 1$ matrix. $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

- A **row vector** is a $1 \times n$ matrix. $[1 \ 2 \ 3]$

- Zero Matrix** ($\mathbf{0}_{m \times n}$): All entries are $\mathbf{0}$. $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

- Identity Matrix** ($\mathbf{I}_n$): A diagonal matrix where all diagonal entries are $\mathbf{1}$. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

- Triangular Matrices:**

- Upper Triangular:** Entries below the main diagonal are $\mathbf{0}$. $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$

- Lower Triangular:** Entries above the main diagonal are $\mathbf{0}$. $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{bmatrix}$

- Strictly Triangular:** Entries on the main diagonal are also $\mathbf{0}$. $\begin{bmatrix} 0 & 2 & 3 \\ 0 & 0 & 5 \\ 0 & 0 & 0 \end{bmatrix}$

- Symmetric Matrix:** A matrix where $\mathbf{A} = \mathbf{A}^T$ (i.e., $a_{ij} = a_{ji}$ for all i, j). $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$

2. Matrix Operations

Equality

- Two matrices $\mathbf{A}$ and $\mathbf{B}$ are equal if:
 - They have the same size.
 - Their corresponding elements are equal.

Matrix Addition & Scalar Multiplication

- Addition:** Sum of corresponding entries. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$
- Scalar Multiplication:** Each entry is multiplied by a scalar $\mathbf{c}$. $3 \times \begin{bmatrix} 1 & -2 \\ -3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & -6 \\ -9 & 12 \end{bmatrix}$

Matrix Multiplication

- If $\mathbf{A}$ is an $\mathbf{m} \times \mathbf{p}$ matrix and $\mathbf{B}$ is a $\mathbf{p} \times \mathbf{n}$ matrix, their product $\mathbf{AB}$ is an $\mathbf{m} \times \mathbf{n}$ matrix.
 $(AB)_{ij} = \sum_{k=1}^p a_{ik}b_{kj}$

Example:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \times \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} (1)(5) + (2)(7) & (1)(6) + (2)(8) \\ (3)(5) + (4)(7) & (3)(6) + (4)(8) \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

MATLAB Code:

```
A = [1 2; 3 4];
B = [5 6; 7 8];
C = A * B; % Matrix multiplication
```

Transpose of a Matrix

- The **transpose** of a matrix $\mathbf{A}$, denoted $\mathbf{A}^T$, flips rows into columns.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

3. Linear Systems and Matrix Equations

- A system of $\mathbf{m}$ linear equations in $\mathbf{n}$ variables can be written as a matrix equation: $Ax = b$
- **Homogeneous System:** A system is **homogeneous** if $\mathbf{b} = \mathbf{0}$.

Example:

Solve the system:

$$\begin{cases} 2x + 3y = 5 \\ 4x + 6y = 10 \end{cases}$$

MATLAB Code:

```
A = [2 3; 4 6];
b = [5; 10];
x = rref([A b]);
```

4. Determinants

Determinant of a 2x2 Matrix

For: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ The determinant is: $\det(A) = ad - bc$

Determinant of a 3x3 Matrix

For:

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

The determinant is:

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Determinant by Row Reduction

- Row operations affect determinants:
 1. Swapping rows negates the determinant.

2. Multiplying a row by a scalar multiplies the determinant by that scalar.
3. Adding a multiple of one row to another does not change the determinant.

5. Matrix Inverses and Invertibility

Inverse of a 2x2 Matrix

For: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ If $ad - bc \neq 0$, then: $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Inverse of Elementary Matrices

- The inverse of an elementary matrix is also an elementary matrix.

Equivalent Statements for Invertibility

Let A be an $n \times n$ matrix. The following are equivalent:

1. A is invertible.
2. A^T is invertible.
3. There exists a matrix B such that $BA = I$ (left inverse).
4. There exists a matrix B such that $AB = I$ (right inverse).
5. The reduced row-echelon form of A is the identity matrix.
6. A can be expressed as a product of elementary matrices.
7. The homogeneous system $Ax = 0$ has only the trivial solution.
8. For any b , the system $Ax = b$ has a unique solution.

MATLAB Code for Determinants and Inverses:

```
A = [1 2; 3 4];
detA = det(A); % Compute determinant
invA = inv(A); % Compute inverse
```

Properties of Matrices

Properties of Matrix Addition and Scalar Multiplication

- **Commutativity:** $A + B = B + A$ for any matrices A, B of the same size.
- **Associativity:** $(A + B) + C = A + (B + C)$.
- **Additive Identity:** There exists a zero matrix 0 such that $A + 0 = A$ for any matrix A .
- **Additive Inverse:** For every matrix A , there exists $-A$ such that $A + (-A) = 0$.
- **Distributive Property:**
 - $c(A + B) = cA + cB$ for any scalar c .
 - $(c + d)A = cA + dA$ for any scalars c, d .
 - $c(dA) = (cd)A$ for any scalars c, d .
- **Multiplicative Identity for Scalars:** $1A = A$ for any matrix A .
- if $Ax = b$, where b is a matrix. Then, $x = A^{-1}b$

Properties of Matrix Multiplication

- **Associativity:** $(AB)C = A(BC)$ for any matrices A, B, C with compatible dimensions.
- **Distributive Property:**
 - $A(B + C) = AB + AC$.
 - $(A + B)C = AC + BC$.
- **Multiplicative Identity:** For an $n \times n$ identity matrix I_n , we have:
 - $AI_n = A$ and $I_nA = A$ for any $n \times n$ matrix A .
- **Non-Commutativity:** In general, $AB \neq BA$ (matrix multiplication is not commutative).
- **Multiplication by Zero Matrix:** $A0 = 0A = 0$ for any matrix A .

Properties of Transpose

- **Transpose of a Sum:** $(A + B)^T = A^T + B^T$.
- **Transpose of a Product:** $(AB)^T = B^T A^T$.
- **Transpose of a Scalar Multiple:** $(cA)^T = cA^T$ for any scalar c .
- **Double Transpose:** $(A^T)^T = A$.
- **Transpose of Identity Matrix:** $I_n^T = I_n$.

Properties of Homogeneous Linear System

- **Trivial Solution:** If $Ax = 0$, then $x = 0$ is always a solution.
- **Existence of Non-Trivial Solutions:** If A is $m \times n$ with $n > m$, then there exist infinitely many solutions.
- **Relation to Null Space:** The set of all solutions forms a vector space called the **null space** of A .
- **Scaling Property:** If x is a solution, then cx is also a solution for any scalar c .

Properties of Inverses

- **Uniqueness:** If A^{-1} exists, it is unique.
- **Inverse of a Product:** $(AB)^{-1} = B^{-1}A^{-1}$ for invertible matrices A, B .
- **Inverse of a Transpose:** $(A^T)^{-1} = (A^{-1})^T$ for invertible A .
- **Inverse of an Identity Matrix:** $I_n^{-1} = I_n$.
- **Inverse of a Scalar Multiple:** If A is invertible, then $(cA)^{-1} = \frac{1}{c}A^{-1}$ for any nonzero scalar c .

Properties of Determinant

- **Determinant and Invertibility:** A matrix is invertible if and only if its determinant is nonzero.
- **Multiplicative Property:** $\det(AB) = \det(A) \cdot \det(B)$ for square matrices A, B .
- **Effect of Row Operations:**
 - Swapping two rows negates the determinant.
 - Multiplying a row by c multiplies the determinant by c .
 - Adding a multiple of one row to another does not change the determinant.
- **Determinant of an Identity Matrix:** $\det(I_n) = 1$.
- **Determinant of a Triangular Matrix:** The determinant is the product of the diagonal entries.
- **Inverse and Determinant:** If A is invertible, then $\det(A^{-1}) = \frac{1}{\det(A)}$.
- **Determinant of a Scalar Multiple:** $\det(cA) = c^n \det(A)$ for an $n \times n$ matrix A .